

§ Principle of Inclusion-Exclusion (PIE) Let $\mathcal{U}$ be a universal set for a counting problem. Let $C_1, C_2, \dots, C_n$ be a finite collection of finite sets where each $C_i \subseteq \mathcal{U}$. The intersection of all C_i defines a condition set for the cardinality, C , as follows.

$$C = \bigcap_{i \in [n]} C_i = C_1 \cap C_2 \cap \dots \cap C_n$$

and the complementary sets $\overline{C_i} = \mathcal{U} \setminus C_i$ for $i \in [n] = \{1, 2, \dots, n\}$ define the sets of elements that do not satisfy the collection of conditions, C . Then, the number of elements in $\mathcal{U}$ that satisfy all the conditions in C is given as follows.

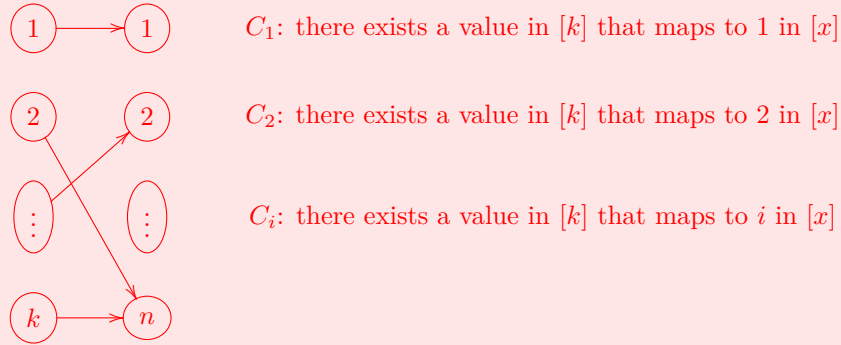
$$\begin{aligned} |C| &= \left| \bigcap_{i=1}^n C_i \right| = |U| - \left| \bigcup_{i=1}^n \overline{C_i} \right| = \sum_{I \subseteq [n]} (-1)^{|I|} \left| \bigcap_{i \in I} \overline{C_i} \right| \\ &= |U| - \sum_{i=1}^n |\overline{C_i}| + \sum_{1 \leq i < j \leq n} |\overline{C_i} \cap \overline{C_j}| - \dots + (-1)^n |\overline{C_1} \cap \overline{C_2} \cap \dots \cap \overline{C_n}| \end{aligned}$$

where the intersection over an empty index set, $\bigcap_{i \in \emptyset} \overline{C_i}$, is defined as the universal set U .

§ Cardinality of Surjective Functions The number of surjective functions from a set $[k] = \{1, 2, \dots, k\}$ to a set $[n] = \{1, 2, \dots, n\}$ is given by the formula:

$$\mathcal{S}(k, n) = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^k$$

Proof: By the definition of surjective functions, a function $f : [k] \rightarrow [n]$ is surjective if and only if for every $i \in [n]$, there exists at least one $j \in [k]$ such that $f(j) = i$. Let C_i be defined as follows.



Then, $|C|$ is the number of surjective functions by definition. Then, by Principle of Inclusion-Exclusion (PIE), the number of surjective functions is given as follows.

$$\begin{aligned} |C| &= n^k - \binom{n}{1} (n-1)^k + \binom{n}{2} (n-2)^k - \dots + (-1)^n \binom{n}{n} (n-n)^k \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^k \end{aligned}$$

since $\sum_{i=1}^n |\overline{C_i}|$ is the number of functions from $[k]$ to $[n]$ such that a single value, i , is not in the range for each $i \in [n]$, which is given by $\binom{n}{1} (n-1)^k$. With similar reasoning, $\sum_{1 \leq i < j \leq n} |\overline{C_i} \cap \overline{C_j}| = \binom{n}{2} (n-2)^k$ and so on.

■

Remark: The following definitions (or results) are already proved in class or on previous ICA, Wbwk, Hmwk assignments, and in Sections 4.4, 4.5, and 4.6 of the textbook.

§ Set Equality Let A and B be sets. The sets A and B are equal ($A = B$) iff $A \subseteq B$ and $B \subseteq A$.

§ Subset Relation Let A and B be sets. A is a subset of B ($A \subseteq B$) iff for all $x \in A$ only if $x \in B$.

§ Union of Sets Let A and B be sets. The union of A and B , denoted $A \cup B$, is the set of all elements that are in A , or in B , or in both. That is, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

§ Intersection of Sets Let A and B be sets. The intersection of A and B , denoted $A \cap B$, is the set of all elements that are in both A and B . That is, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

§ Set Difference Let A, B be subsets of a universal set $\mathcal{U}$. The set difference $A \setminus B$ is defined to be the set $\{x \in \mathcal{U} \mid x \in A \text{ and } x \notin B\}$.

It is sometimes called the relative complement of B in A , and denoted in the textbook as $A - B$.

§ Power Set Let A be a set. The power set of A , denoted $\mathcal{P}(A)$ or $2^{|A|}$, is the set of all subsets of A . That is, $\mathcal{P}(A) = \{S \in \mathcal{U} : S \subseteq A\}$.

§ Complementary Set Let A be a subset of the universal set $\mathcal{U}$. The complement of A , denoted A^c or $\overline{A}$, is the set of all elements in $\mathcal{U}$ that are not in A . That is, $\overline{A} = \{x \in \mathcal{U} : x \notin A\}$.

§ Cartesian Product Let A and B be sets. The Cartesian product of A and B , denoted $A \times B$, is the set consisting of all ordered pairs:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

§ Pairwise Disjoint Sets Let $A_1, A_2, \dots, A_n$ be sets. These sets are **pairwise disjoint** if for all $i \neq j$, we have $A_i \cap A_j = \emptyset$.

Remark: The following definitions (or results) are already proved in class or on previous ICA, Wbwk, Hmwk assignments, and contents of the textbook.

§ Binary Relation Let A and B be sets. A binary relation R from A to B is a subset of $A \times B$. If (and only if)

$$(a, b) \in R \subseteq A \times B$$

, then we say that a is related to b and write $a R b$. Otherwise, $a \not R b$.

§ Digraph A digraph (or directed graph) is a set of vertices V together with a set of edges $E \subseteq V \times V$.

§ Bipartite Digraph A digraph is bipartite if (and only if) there exist subsets $A, B \subseteq V$ such that

1. $V = A \cup B$ where $A \cap B = \emptyset$ and $A, B \neq \emptyset$.
2. for every $e \in E, e = (a, b)$ for some $a \in A, b \in B$.

Example: Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $R = \{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B . Binary relations on finite sets like this can be visualized as a bipartite digraph:

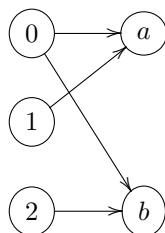


Figure 1: A bipartite graph example

§ Inverse Relation Let A and B be sets. Let R be a relation from A to B . The inverse relation R^{-1} from B to A is the set of ordered pairs $\{(b, a) \mid (a, b) \in R\}$.

§ Relation on a Set A relation on a set A is a relation from A to A .

§ Relation Properties Let A be a set and R a relation on A . Relations on a set are classified by their properties. The most important properties are:

1. **Reflexive:** The relation R is reflexive if (and only if) for all $x \in A$, $(x, x) \in R$.
2. **Symmetric:** The relation R is symmetric if (and only if) for all $x, y \in A$, $(x, y) \in R$ implies $(y, x) \in R$.
3. **Antisymmetric:** The relation R is antisymmetric if (and only if) for all $x, y \in A$, $(x, y) \in R$ and $(y, x) \in R$ only if $x = y$.
4. **Asymmetric:** The relation R is asymmetric if (and only if) for all $x, y \in A$, if $(x, y) \in R$, then $(y, x) \notin R$.
5. **Transitive:** The relation R is transitive if (and only if) for all $x, y, z \in A$, if $(x, y) \in R$ and $(y, z) \in R$, then $(x, z) \in R$.

§ Equivalence Classes Form Partition Let R be an equivalence relation on a nonempty set A . Then $P = \{[a]_R \mid a \in A\}$, the set of equivalence classes resulting from R , is a partition of A .

§ Partition Induces Equivalence Relation Let A be a nonempty set and $P = \{S_i \mid i \in I\}$ a partition of A . Then there exists an equivalence relation R which has the sets $S_i, i \in I$, as its equivalence classes.

Remark: Note that the set I above is an arbitrary index set. It is not necessarily finite, countable, etc!

§ Modulo Operation Let $a, m \in \mathbb{Z}$ with m positive. The remainder when a is divided by m is denoted $a \bmod m$.

§ Congruence Modulo Let $a, b, m \in \mathbb{Z}$ with m positive. If $m \mid (a - b)$ then we say that a is congruent to b modulo m and write $a \equiv b \pmod{m}$.

Example: Let R be an equivalence relation on the set $\{1, 2, 3, 4, 5, 6, 7\}$ such that $1R3, 3R4, 4R7$, and $2R6$. Suppose that R has three equivalence classes.

Example: Let S be another equivalence relation on the set $\{1, 2, 3, 4, 5, 6\}$. Suppose the equivalence classes of S are $\{1, 4, 5\}$, $\{2, 6\}$, and $\{3\}$.

Example: Let $[a]_n$ denote the congruence class of $a \in \mathbb{Z}$ modulo n (that is, the equivalence class of a under the relation congruence modulo n) for integers $n \geq 2$. Are the following propositions true or false?

§ Equivalence Relation Let A be a nonempty set. A relation R on A is an equivalence relation if and only if R is reflexive, symmetric, and transitive.

§ Partition Let A be a nonempty set. A partition of A is a collection of nonempty, pairwise disjoint subsets of A whose union is A .

§ Partition Refinement Let A be a nonempty set. A partition P_1 of A is a refinement of a partition P_2 of A if and only if every set in P_1 is a subset of some set in P_2 .

§ Equivalence Class Let A be a nonempty set and R an equivalence relation on A . For each $a \in A$, the equivalence class of a with respect to R is the set $[a]_R = \{b \in A \mid aRb\}$.

§ Irreflexive Relation Let A be a nonempty set. A relation R on A is irreflexive if and only if for every $a \in A$, $(a, a) \notin R$.

§ Diagonal Relation Let A be a nonempty set. The diagonal relation on A is the set $\Delta = \{(a, a) \mid a \in A\}$.

Lecture 5: Dirichlet's Theorem and applications of the FTA

§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions) Let $a, b \in \mathbb{Z}$ with $b > 0$ and $\gcd(a, b) = 1$. Then the arithmetic progression

$$a, a + b, a + 2b, a + 3b, \dots$$

contains infinitely many prime numbers.

§ Remark (on Theorem Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions)) Setting $a = 1$ and $b = 1$ recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4) There are infinitely many primes of the form $4n + 3$, where n is a nonnegative integer.

§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4) Let $a, b \in \mathbb{Z}$. If $a \equiv 1 \pmod{4}$ and $b \equiv 1 \pmod{4}$, then $ab \equiv 1 \pmod{4}$.

Proof. Write $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so $ab \equiv 1 \pmod{4}$. This proves the lemma. $\square$

Proof of Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4). Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3, P_2, P_3, \dots, P_r,$$

all distinct and each $P_i \equiv 3 \pmod{4}$.

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that $N \equiv -1 \equiv 3 \pmod{4}$.

Let q be any prime divisor of N . We make two observations:

1. For each $i = 1, \dots, r$ we have $N \equiv -1 \pmod{P_i}$, so $P_i \nmid N$. Thus q is not equal to any of the P_i .
2. Since $N \equiv 3 \pmod{4}$, it is not possible that every prime divisor of N is congruent to 1 modulo 4. Indeed, a product of integers each congruent to 1 modulo 4 is itself congruent to 1 modulo 4 by Lemma Lemma 1.23 (Product of numbers congruent to 1 modulo 4), contradicting $N \equiv 3 \pmod{4}$. Hence some prime divisor q of N satisfies $q \equiv 3 \pmod{4}$.

Combining the two observations, q is a prime congruent to 3 modulo 4 that is not among $P_1, \dots, P_r$, contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4. $\square$

§ Remark (on Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4)) The method above—building an integer with residue 3 modulo 4 whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and L -series).

Congruences

§ Historical remark (Gauss and congruences) Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

§ Definition (Congruence modulo m) Let $a, b, m \in \mathbb{Z}$ with $m > 0$. We say that a is congruent to b modulo m , and write

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$. The integer m is called the *modulus* of the congruence.

§ Remark (notation) The notation $a \not\equiv b \pmod{m}$ means $m \nmid (a - b)$. The modulus is customarily taken to be positive; we will assume $m > 0$ in the sequel.

§ Proposition 2.1 (Congruence modulo m is an equivalence relation) Congruence modulo m is reflexive, symmetric, and transitive on $\mathbb{Z}$; hence it is an equivalence relation.

Proof. Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$ and $m \mid 0$, so $a \equiv a \pmod{m}$.

Symmetric: If $a \equiv b \pmod{m}$ then $m \mid (a - b)$, hence $m \mid -(a - b) = (b - a)$, so $b \equiv a \pmod{m}$.

Transitive: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$ then $m \mid (a - b)$ and $m \mid (b - c)$, therefore $m \mid (a - b) + (b - c) = a - c$, so $a \equiv c \pmod{m}$. $\square$

§ Consequence 2.2 (Partition into residue classes) The set $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo m .

§ Definition (Notation for residue class) For $x \in \mathbb{Z}$, write $[x]_m$ (or simply $[x]$ when m is understood) for the equivalence class of x modulo m :

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse $[x]_m$ with the floor function.)

Example: The equivalence classes modulo 4

$$[0]_4 = \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},$$

$$[1]_4 = \{x : x = 4n + 1, n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},$$

$$[2]_4 = \{x : x = 4n + 2, n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},$$

$$[3]_4 = \{x : x = 4n + 3, n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

Thus $\mathbb{Z}$ is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of 0, 1, 2, 3 modulo 4.

§ Definition (Complete residue system modulo m) A set $R \subset \mathbb{Z}$ is a *complete residue system modulo m* if every integer is congruent modulo m to exactly one element of R .

§ Example (complete residue systems) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m . Also, for example, $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4, since $6 \equiv 2$, $-11 \equiv 1$, $19 \equiv 3$, and $1988 \equiv 0 \pmod{4}$.

§ Proposition 2.3 (Standard complete residue system) The set $\{0, 1, 2, \dots, m - 1\}$ is a complete residue system modulo m .

Proof. Let $a \in \mathbb{Z}$. By the Division Algorithm there exist unique $q, r \in \mathbb{Z}$ with $0 \leq r < m$ such that $a = mq + r$. Then $a - r = mq$, so $m \mid (a - r)$ and $a \equiv r \pmod{m}$, with $r \in \{0, 1, \dots, m - 1\}$. This shows every integer is congruent to at least one element of $\{0, 1, \dots, m - 1\}$.

To show uniqueness: suppose $r_1, r_2 \in \{0, 1, \dots, m-1\}$ and $r_1 \equiv r_2 \pmod{m}$. Then $m \mid (r_1 - r_2)$, but $-(m-1) \leq r_1 - r_2 \leq m-1$. The only multiple of m in this range is 0, so $r_1 - r_2 = 0$ and hence $r_1 = r_2$. Thus each integer is congruent to exactly one element of $\{0, 1, \dots, m-1\}$. $\square$

§ Proposition 2.4 (Arithmetic with congruences) Let $a, b, c, d \in \mathbb{Z}$ and $m > 0$. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

1. $a + c \equiv b + d \pmod{m}$,
2. $ac \equiv bd \pmod{m}$.

Proof. From $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ we have $m \mid (a - b)$ and $m \mid (c - d)$.

For (1): $m \mid (a - b) + (c - d) = (a + c) - (b + d)$, hence $a + c \equiv b + d \pmod{m}$.

For (2): Note $m \mid (a - b)$ implies $m \mid c(a - b)$, and $m \mid (c - d)$ implies $m \mid b(c - d)$. Add these divisibilities to get $m \mid c(a - b) + b(c - d) = ac - bd$, so $ac \equiv bd \pmod{m}$. $\square$

L6: Congruences

§ Remark (Proposition 2.4 (Arithmetic with congruences) in class terms) Recall that Proposition 2.4 (Arithmetic with congruences) from the previous lecture can be expressed in terms of equivalence classes modulo m : for residue classes $[a], [c] \in \mathbb{Z}_m$,

$$[a] + [c] = [a + c] \quad \text{and} \quad [a][c] = [ac].$$

This viewpoint allows construction of addition and multiplication tables for residue classes (modulo m) using any complete residue system.

Examples: tables for $\mathbb{Z}_4$ and $\mathbb{Z}_5$

+	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[1]_4$	$[1]_4$	$[2]_4$	$[3]_4$	$[0]_4$
$[2]_4$	$[2]_4$	$[3]_4$	$[0]_4$	$[1]_4$
$[3]_4$	$[3]_4$	$[0]_4$	$[1]_4$	$[2]_4$

·	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$	$[0]_4$
$[1]_4$	$[0]_4$	$[1]_4$	$[2]_4$	$[3]_4$
$[2]_4$	$[0]_4$	$[2]_4$	$[0]_4$	$[2]_4$
$[3]_4$	$[0]_4$	$[3]_4$	$[2]_4$	$[1]_4$

§ Remark (structure of $\mathbb{Z}_4$) Set $\mathbb{Z}_4 := \{[0]_4, [1]_4, [2]_4, [3]_4\}$. With the above addition and multiplication it is a ring, but not a field: not every nonzero element has a multiplicative inverse (e.g., $[2]_4$ is a zero divisor).

+	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[1]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$
$[2]_5$	$[2]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$
$[3]_5$	$[3]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$
$[4]_5$	$[4]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$

·	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$	$[0]_5$
$[1]_5$	$[0]_5$	$[1]_5$	$[2]_5$	$[3]_5$	$[4]_5$
$[2]_5$	$[0]_5$	$[2]_5$	$[4]_5$	$[1]_5$	$[3]_5$
$[3]_5$	$[0]_5$	$[3]_5$	$[1]_5$	$[4]_5$	$[2]_5$
$[4]_5$	$[0]_5$	$[4]_5$	$[3]_5$	$[2]_5$	$[1]_5$

§ Remark (structure of $\mathbb{Z}_5$) Let $\mathbb{Z}_5 := \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$. Because 5 is prime every nonzero class has a multiplicative inverse, so $\mathbb{Z}_5$ is a field.

§ Question (why extra structure?) Why does $\mathbb{Z}_5$ have more structure (a field) than $\mathbb{Z}_4$ (only a ring)?

§ Answer (primality of modulus) Essentially: $\mathbb{Z}_m$ is a field if and only if m is prime. If m is composite, then some nonzero classes are zero divisors (no inverses). If $m = p$ is prime, then every nonzero class has an inverse (standard proof via Bezout's identity).

Cancellation and a useful proposition

§ Proposition 2.5 (Cancellation in congruences) Let $a, b, c, m \in \mathbb{Z}$ with $m > 0$, and let $d = \gcd(c, m)$. Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/d}.$$

Proof. Suppose $ca \equiv cb \pmod{m}$, i.e. $m \mid c(a - b)$. Let $d = \gcd(c, m)$. Write $c = d \cdot c'$ and $m = d \cdot m'$ so that $\gcd(c', m') = 1$. From $m \mid c(a - b)$ we get $dm' \mid dc'(a - b)$, hence $m' \mid c'(a - b)$. Since $\gcd(c', m') = 1$, it follows that $m' \mid (a - b)$, i.e. $a \equiv b \pmod{m'}$, which is $a \equiv b \pmod{m/d}$.

Conversely, if $a \equiv b \pmod{m/d}$ then $m/d \mid (a - b)$, so $m/d \cdot d = m \mid d(a - b)$. But $d \mid c$, so $c(a - b)$ is a multiple of $d(a - b)$, hence $m \mid c(a - b)$ and $ca \equiv cb \pmod{m}$. This proves the equivalence. $\square$

Linear congruences in one variable

§ Definition (linear congruence) A congruence of the form

$$ax \equiv b \pmod{m},$$

with unknown $x \in \mathbb{Z}$ and given integers a, b, m with $m > 0$, is called a *linear congruence* in the variable x .

§ Examples (small checks)

- $2x \equiv 3 \pmod{4}$ has no solution: checking $x \equiv 0, 1, 2, 3 \pmod{4}$ yields no solution.
- $2x \equiv 3 \pmod{5}$ has the unique solution $x \equiv 4 \pmod{5}$.
- $2x \equiv 4 \pmod{6}$ has two incongruent solutions modulo 6: $x \equiv 2, 5 \pmod{6}$.
- $3x \equiv 9 \pmod{6}$ has three incongruent solutions modulo 6: $x \equiv 1, 3, 5 \pmod{6}$.

§ Theorem 2.6 (Solvability and number of solutions of linear congruences) Let $a, b, m \in \mathbb{Z}$ with $m > 0$ and let $d = \gcd(a, m)$. Then:

1. The congruence $ax \equiv b \pmod{m}$ has a solution iff $d \mid b$.
2. If $d \mid b$ then there are exactly d incongruent solutions modulo m . Concretely, if x_0 is one solution, then all solutions are

$$x = x_0 + \frac{m}{d}n, \quad n \in \mathbb{Z},$$

and the classes $x_0 + \frac{m}{d}n$ for $n = 0, 1, \dots, d-1$ give a complete set of incongruent solutions modulo m .

Proof. We use the standard equivalence between the congruence and a linear Diophantine equation. The congruence $ax \equiv b \pmod{m}$ is equivalent to the existence of an integer y such that

$$ax - my = b. \quad (*)$$

Thus solvability of the congruence is equivalent to solvability of the linear Diophantine equation (*).

(1) If (*) has an integer solution, then $d = \gcd(a, m)$ divides the left-hand side $ax - my$ and hence divides b . Conversely, if $d \mid b$, write $b = de$ for some $e \in \mathbb{Z}$. By Bezout's identity (see ?? or the standard result), there exist integers r, s with $ar + ms = d$. Multiply both sides by e to obtain $are + mse = de = b$. Then $x_0 = re$ and $y_0 = -se$ solve (*), so a solution exists.

(2) Fix any solution x_0 of $ax \equiv b \pmod{m}$. For any integer n consider

$$x = x_0 + \frac{m}{d}n.$$

Then

$$ax = ax_0 + a \frac{m}{d}n = ax_0 + \frac{a}{d}mn \equiv ax_0 \equiv b \pmod{m},$$

so each such x is a solution. Thus there are infinitely many integer solutions, all congruent modulo $\frac{m}{d}$ shifts.

Now suppose x is any solution. Then subtracting the equation for x_0 from the equation for x yields

$$a(x - x_0) \equiv 0 \pmod{m},$$

i.e. $m \mid a(x - x_0)$. Write $a = da'$ and $m = dm'$, with $\gcd(a', m') = 1$. Then $m'd \mid da'(x - x_0)$, so $m' \mid a'(x - x_0)$. Since $\gcd(a', m') = 1$ we deduce $m' \mid (x - x_0)$, hence $x - x_0 = m'n = \frac{m}{d}n$ for some $n \in \mathbb{Z}$. This shows every solution has the stated form. Two such solutions

$$x_0 + \frac{m}{d}n_1 \quad \text{and} \quad x_0 + \frac{m}{d}n_2$$

are congruent modulo m iff

$$\frac{m}{d}(n_1 - n_2) \equiv 0 \pmod{m} \iff m \mid \frac{m}{d}(n_1 - n_2).$$

Dividing both sides by m/d yields $d \mid (n_1 - n_2)$, so $n_1 \equiv n_2 \pmod{d}$. Therefore choosing $n = 0, 1, \dots, d-1$ produces exactly d incongruent solutions modulo m . $\square$

§ Corollary 2.7 (Explicit solution set) Under the hypotheses of Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences), if $d = \gcd(a, m)$ and $d \mid b$, then the d incongruent solutions modulo m are

$$x_0 + \frac{m}{d}n, \quad n = 0, 1, \dots, d-1,$$

where x_0 is any particular solution.

Worked example

§ Example (solve $16x \equiv 8 \pmod{28}$) We solve $16x \equiv 8 \pmod{28}$. Compute $d = \gcd(16, 28)$.

Euclidean algorithm:

$$28 = 1 \cdot 16 + 12, \quad 16 = 1 \cdot 12 + 4, \quad 12 = 3 \cdot 4 + 0,$$

so $d = 4$. Since $4 \mid 8$, the congruence has $d = 4$ incongruent solutions modulo 28 by Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences).

One method: divide the congruence by $d = 4$ to get (working modulo $28/4 = 7$)

$$4x \equiv 2 \pmod{7}.$$

Since $4^{-1} \equiv 2 \pmod{7}$ (because $4 \cdot 2 = 8 \equiv 1 \pmod{7}$), multiply both sides by 2:

$$x \equiv 4 \pmod{7}.$$

Thus the solutions modulo 28 are

$$x \equiv 4 + 7n, \quad n = 0, 1, 2, 3,$$

which gives the incongruent solutions $x \equiv 4, 11, 18, 25 \pmod{28}$.

(Alternatively, one can find an explicit x_0 via the extended Euclidean computation: from $4 = 2 \cdot 16 - 1 \cdot 28$ we get $8 = 4 \cdot 16 - 2 \cdot 28$, so $x_0 = 4$ is a particular solution, yielding the same solution set.)

Multiplicative inverses

§ Definition (multiplicative inverse modulo m) If $a \in \mathbb{Z}$ and there exists $x \in \mathbb{Z}$ with $ax \equiv 1 \pmod{m}$, then x is called a *multiplicative inverse* of a modulo m (often denoted $a^{-1} \pmod{m}$).

§ Corollary 2.8 (When inverses exist) The congruence $ax \equiv 1 \pmod{m}$ has a solution iff $\gcd(a, m) = 1$. In that case the inverse is unique modulo m .

Proof. Apply Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences) with $b = 1$. If $d = \gcd(a, m) > 1$ then $d \nmid 1$ so there is no solution. If $d = 1$ then there is exactly one solution modulo m . $\square$

§ Example (solve $5x \equiv 12 \pmod{16}$) Since $\gcd(5, 16) = 1$, the congruence has a unique solution modulo 16 by Corollary Corollary 2.8 (When inverses exist). One checks that $5^{-1} \pmod{16} = 13$ because $5 \cdot 13 = 65 \equiv 1 \pmod{16}$. Multiplying by 13 gives

$$x \equiv 13 \cdot 12 \pmod{16}.$$

Compute $13 \cdot 12 = 156 \equiv 12 \pmod{16}$ (since $156 - 144 = 12$), so $x \equiv 12 \pmod{16}$.

L7: The Chinese Remainder Theorem, Wilson's Theorem, Fermat's Little Theorem

Example (motivation). Find a positive integer x with

$$x \equiv 2 \pmod{3}, \quad x \equiv 1 \pmod{4}, \quad x \equiv 3 \pmod{5}.$$

This is a system of linear congruences in one variable.

§ Theorem 2.9 (Chinese Remainder Theorem) Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers, and let $b_1, \dots, b_n$ be arbitrary integers. Then the system

$$x \equiv b_i \pmod{m_i} \quad (i = 1, \dots, n)$$

has a solution, and any two solutions are congruent modulo $M := m_1 m_2 \cdots m_n$. Hence the solution is unique modulo M .

Proof. Let $M := \prod_{i=1}^n m_i$ and define $M_i := M/m_i$. Since $(M_i, m_i) = 1$ by the pairwise coprimality hypothesis, by Corollary 2.8 (When inverses exist) (existence of inverses when gcd is 1) there exists x_i with

$$M_i x_i \equiv 1 \pmod{m_i}.$$

Put

$$x := \sum_{i=1}^n b_i M_i x_i.$$

For a fixed index i , reduce modulo m_i : every term $b_j M_j x_j$ with $j \neq i$ is a multiple of m_i (since $m_i \mid M_j$ for $j \neq i$), while $M_i x_i \equiv 1 \pmod{m_i}$ by construction. Thus $x \equiv b_i \pmod{m_i}$ for each i , so x is a solution.

If x' is another solution, then $m_i \mid (x - x')$ for every i , and since the m_i are pairwise coprime it follows that $M \mid (x - x')$. Hence $x \equiv x' \pmod{M}$, proving uniqueness modulo M . $\square$

Example (continued). Here $m_1 = 3, m_2 = 4, m_3 = 5$, so $M = 60$ and

$$M_1 = 60/3 = 20, \quad M_2 = 60/4 = 15, \quad M_3 = 60/5 = 12.$$

Solve $M_i x_i \equiv 1 \pmod{m_i}$:

$$20x_1 \equiv 1 \pmod{3} \implies 2x_1 \equiv 1 \pmod{3} \implies x_1 \equiv 2 \pmod{3},$$

so choose $x_1 = 2$.

$$15x_2 \equiv 1 \pmod{4} \implies 3x_2 \equiv 1 \pmod{4} \implies x_2 \equiv 3 \pmod{4},$$

so choose $x_2 = 3$.

$$12x_3 \equiv 1 \pmod{5} \implies 2x_3 \equiv 1 \pmod{5} \implies x_3 \equiv 3 \pmod{5},$$

so choose $x_3 = 3$.

Now

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 = 80 + 45 + 108 = 233.$$

Hence $x \equiv 233 \equiv 53 \pmod{60}$, and the least positive solution is 53.

§ Lemma 2.10 (Self-inverse modulo a prime) Let p be a prime and $a \in \mathbb{Z}$. Then a is its own inverse modulo p (i.e. $a^2 \equiv 1 \pmod{p}$) if and only if $a \equiv \pm 1 \pmod{p}$.

Proof. If $a^2 \equiv 1 \pmod{p}$ then $p \mid (a-1)(a+1)$. Since p is prime, $p \mid (a-1)$ or $p \mid (a+1)$, so $a \equiv 1$ or $a \equiv -1 \pmod{p}$. The converse is immediate. $\square$

§ Theorem 2.11 (Wilson's Theorem) Let p be a prime. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. The result is trivial for $p = 2, 3$, so assume $p \geq 5$. The integers $1, 2, \dots, p-1$ can be paired into multiplicative inverses modulo p . By Lemma , the only elements that are their own inverses are 1 and $p-1$. Thus, aside from 1 and $p-1$, the remaining $p-3$ elements pair off, and each pair multiplies to 1 modulo p . Therefore

$$(p-1)! \equiv 1 \cdot (\text{product of inverse-pairs}) \cdot (p-1) \equiv 1 \cdot 1 \cdots 1 \cdot (-1) \equiv -1 \pmod{p},$$

as required. $\square$

§ Remark (converse) The converse is also true: if $n > 1$ and $(n-1)! \equiv -1 \pmod{n}$ then n must be prime.

§ Proposition 2.12 (Converse to Wilson's Theorem) Let $n > 1$ be an integer. If $(n-1)! \equiv -1 \pmod{n}$ then n is prime.

Proof. Suppose n is composite, so $n = ab$ with $1 < a \leq b < n$. Then $a \mid (n-1)!$ because $(n-1)!$ contains the factor a , and hence $(n-1)! \equiv 0 \pmod{a}$. The hypothesis $(n-1)! \equiv -1 \pmod{n}$ implies $(n-1)! \equiv -1 \pmod{a}$ (since $a \mid n$), so $-1 \equiv 0 \pmod{a}$, which is impossible. Therefore no such nontrivial factor a exists and n must be prime. $\square$

§ Remark (Wilson primes) A prime p is called a *Wilson prime* if $(p-1)! \equiv -1 \pmod{p^2}$. Only three Wilson primes are known: 5, 13, 563. It is unknown whether infinitely many (or only finitely many) Wilson primes exist.

§ Theorem 2.13 (Fermat's Little Theorem) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the $p-1$ integers $a, 2a, \dots, (p-1)a$. None of these is congruent to 0 $\pmod{p}$ (since $p \nmid a$), and no two are congruent modulo p (if $ia \equiv ja \pmod{p}$ with $1 \leq i < j \leq p-1$, multiply by the inverse of a modulo p , which exists by Corollary 2.8 (When inverses exist), to deduce $i \equiv j \pmod{p}$, impossible). Hence the residues of $a, 2a, \dots, (p-1)a$ modulo p are a permutation of $1, 2, \dots, p-1$. Multiplying all these congruences yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since $p \nmid (p-1)!$ (true for prime p), we may cancel $(p-1)!$ from both sides to obtain $a^{p-1} \equiv 1 \pmod{p}$. (Note: this cancellation step uses only that $\gcd((p-1)!, p) = 1$; Wilson's Theorem is not required.) $\square$

L8: Corollaries to Fermat's Little Theorem, Pseudoprimes, Euler's Theorem, Arithmetic Functions

§ Corollary 2.14 (Inverse modulo a prime) Let p be a prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Then a^{p-2} is the multiplicative inverse of a modulo p ; that is,

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

Proof. By Fermat's Little Theorem (Theorem), $a^{p-1} \equiv 1 \pmod{p}$. Multiply both sides by a^{-1} if you like, or simply rewrite $a^{p-1} = a \cdot a^{p-2}$ to get $a \cdot a^{p-2} \equiv 1 \pmod{p}$. Thus a^{p-2} is an inverse of a modulo p . $\square$

§ Remark (application of Corollary Corollary 2.14 (Inverse modulo a prime)) Corollary Corollary 2.14 (Inverse modulo a prime) is useful for solving linear congruences $ax \equiv b \pmod{p}$ when p is prime and $p \nmid a$: multiply both sides by a^{p-2} to obtain the solution.

§ Corollary 2.15 (Fresh form of Fermat) Let p be prime and $a \in \mathbb{Z}$. Then

$$a^p \equiv a \pmod{p}.$$

Proof. If $p \mid a$ the congruence is trivial. If $p \nmid a$, multiply Fermat's Little Theorem $a^{p-1} \equiv 1 \pmod{p}$ by a to get $a^p \equiv a \pmod{p}$. $\square$

§ Corollary 2.16 (case $a = 2$) For every prime p ,

$$2^p \equiv 2 \pmod{p}.$$

Proof. Take $a = 2$ in Corollary Corollary 2.15 (Fresh form of Fermat). $\square$

§ Question (converse for base 2) Is the converse of Corollary Corollary 2.16 (case $a = 2$) true? That is, if $n > 1$ and $2^n \equiv 2 \pmod{n}$, must n be prime?

§ Answer and counterexample (341) No. Example: $n = 341 = 11 \cdot 31$ is composite but satisfies $2^{341} \equiv 2 \pmod{341}$. Since $(11, 31) = 1$, it suffices to check modulo 11 and 31.

By Fermat's Little Theorem,

$$2^{10} \equiv 1 \pmod{11} \Rightarrow 2^{341} \equiv (2^{10})^{34} \cdot 2 \equiv 1^{34} \cdot 2 \equiv 2 \pmod{11},$$

and

$$2^{30} \equiv 1 \pmod{31} \Rightarrow 2^{341} \equiv (2^{30})^{11} \cdot 2^{11} \equiv 1^{11} \cdot 2^{11} \pmod{31}.$$

Since $2^5 \equiv -1 \pmod{31}$ (or compute directly), we have $2^{11} \equiv 2 \pmod{31}$, so $2^{341} \equiv 2 \pmod{31}$. Hence $2^{341} \equiv 2 \pmod{341}$ although 341 is composite.

§ Definition (pseudoprime to base 2) A composite integer $n > 1$ is called a (*base-2*) *pseudoprime* if

$$2^n \equiv 2 \pmod{n}.$$

§ Remark (pseudoprimes) 341 is the smallest base-2 pseudoprime. There are infinitely many pseudoprimes and they arise for different bases; pseudoprimes show that the naive converse of Fermat's Little Theorem is false.

§ Definition (Euler's phi-function) For $n \in \mathbb{Z}_{>0}$ the Euler totient function $\phi(n)$ is the number of integers $1 \leq x \leq n$

with $\gcd(x, n) = 1$; equivalently,

$$\phi(n) := \#\{x \in \{1, 2, \dots, n\} : \gcd(x, n) = 1\}.$$

§ Examples (values of ϕ)

$$\phi(12) = 4 \quad (\text{the classes } 1, 5, 7, 11), \quad \phi(14) = 6 \quad (\text{the classes } 1, 3, 5, 9, 11, 13),$$

and for a prime p , $\phi(p) = p - 1$ (classes $1, 2, \dots, p - 1$).

§ Theorem 2.17 (Euler's Theorem) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

Proof. Let $r_1, r_2, \dots, r_{\phi(m)}$ be the positive integers $\leq m$ that are relatively prime to m . Since $\gcd(a, m) = 1$, multiplication by a permutes this reduced residue system: the numbers $ar_1, \dots, ar_{\phi(m)}$ are all distinct modulo m and each is relatively prime to m (see Lemma 1.5 and Lemma 1.14 referenced earlier). Hence the set $\{ar_i \bmod m\}$ is the same set of residues as $\{r_i\}$. Multiplying all residues yields

$$a^{\phi(m)} \prod_{i=1}^{\phi(m)} r_i \equiv \prod_{i=1}^{\phi(m)} r_i \pmod{m}.$$

Since $\gcd(\prod_i r_i, m) = 1$, we may cancel the product to obtain $a^{\phi(m)} \equiv 1 \pmod{m}$. □

§ Remark (special case) Fermat's Little Theorem is the special case $m = p$ prime, since $\phi(p) = p - 1$.

§ Definition (reduced residue system) A set of $\phi(m)$ integers each relatively prime to m and pairwise incongruent modulo m is called a *reduced residue system* modulo m .

§ Example (reduced residue systems) $\{1, 5, 7, 11\}$ is a reduced residue system modulo 12. If a is coprime to m , then $\{ar_1, \dots, ar_{\phi(m)}\}$ is also a reduced residue system modulo m (this was the idea used in the proof of Euler's Theorem).

§ Corollary 2.19 (inverse via Euler) Let $m > 0$ and $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$. Then $a^{\phi(m)-1}$ is the multiplicative inverse of a modulo m .

Proof. From $a^{\phi(m)} \equiv 1 \pmod{m}$ multiply by a^{-1} (or rewrite as $a \cdot a^{\phi(m)-1} \equiv 1$) to see $a^{\phi(m)-1}$ is an inverse of a modulo m . □

§ Remark (use of Corollary Corollary 2.19 (inverse via Euler)) Corollary Corollary 2.19 (inverse via Euler) helps solve linear congruences $ax \equiv b \pmod{m}$ when $\gcd(a, m) = 1$: multiply both sides by $a^{\phi(m)-1}$ to obtain the solution.

§ Definition (arithmetic function) An *arithmetic function* is a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$ (or $\mathbb{R}$, or $\mathbb{Z}$). Examples: the Euler ϕ -function, the divisor-counting function $d(n)$, the sum-of-divisors function $\sigma(n)$, etc.

§ Definition (multiplicative vs completely multiplicative) An arithmetic function f is *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is *completely multiplicative* if $f(mn) = f(m)f(n)$ for all positive integers m, n (no coprimality required).

§ Proposition (values determined by prime powers) If f is multiplicative and

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$$

is the prime factorization of n , then

$$f(n) = \prod_{i=1}^r f(p_i^{a_i}).$$

If f is completely multiplicative, then

$$f(n) = \prod_{i=1}^r f(p_i)^{a_i}.$$

Proof. Write n as a product of relatively prime prime-power factors and apply multiplicativity repeatedly. □

§ Remark (implication) Thus knowing a multiplicative arithmetic function on prime powers (or a completely multiplicative function on primes) determines it everywhere.